Net 5.8 Tbps IM/DD Transmission over 2 km using 25 Simultaneous 100 GHz Comb Channels and a Single SOA

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Abstract: We demonstrate the first 25 channel O-band IM/DD transmission system over 2 km using a single SOA and a 100-GHz-spaced quantum-dot comb laser. We transmitted 3.2 Tbps PAM-4, 4.3 Tbps PAM-6, and 5.76 Tbps PAM-8. © 2026 The Author(s)

1. Introduction

The ever-increasing demand for higher speeds and bandwidths (BW) has driven the adoption of wavelength-division multiplexing (WDM) as a key enabler for scalable optical interconnects. Optical comb lasers have been proposed as novel light sources in high-speed intensity-modulated direct-detection (IM/DD) transmission demonstrations [1, 2]. Among various comb technologies, quantum-dot mode-locked lasers (QD-MLLs) stand out due to their size, efficiency, high per-line output power and compatibility with silicon photonics (SiP) integration [3-5], making them attractive candidates for compact and energy-efficient WDM systems. Nevertheless, QD-MLL-based combs have traditionally been constrained to optical BWs of approximately 8 nm (1.4 THz) with channel spacings of 25 to 100 GHz, restricting achievable symbol rates and requiring many components and amplifiers to scale to multi-Tbps data rates [2, 6-8]. Recent advances have doubled the available optical BWs to 2.3 THz [5] simultaneously supporting high symbol rates and larger channel counts. This progress positions QD-MLL comb sources as strong contenders against alternative multi-wavelength solutions for short-reach, high-speed interconnects [9].

In this paper, we report the first O-band IM/DD optical fibre transmission system employing a QD-MLL with a 100 GHz line spacing to transmit a net data rate 3.2, 4.3, and 5.76 Tbps over 2 km of SMF28 using a single optical amplifier for all channels. This was achieved using 25 comb lines each operating at 70 Gbaud PAM-4 under the 6.7% overhead hard-decision forward error correction (HD-FEC) threshold, 80 Gbaud PAM-6 under the 20% overhead soft-decision FEC (SD-FEC) and 96 Gbaud PAM-8 under the 25% overhead SD-FEC threshold, respectively. We also experimentally investigate the effects of adjusting the laser current and temperature, symbol rate, received optical power (ROP), fibre distances and eye diagrams.

2. Comb Characterization & Experimental Results

The QD-MLL was characterized to optimize the transmission performance by changing its driving current, bias voltage, and temperature using its integrated TEC and electronics. The line spacing is set by the 400 μm long InAs/GaAs QD cavity length and is fixed at 100 GHz. The center wavelength was set to 1312 nm by the drive current and is largely unaffected by the chip temperature as shown in Fig. 1b-c. The number of channels, optical BW, and power per line dependence on comb parameters is also shown, with the bias allowing for ~2 × increase on the number of channels and optical BW. The laser was operated with 225 mA driving current, 4 V bias, and 25°C resulting in 25 lines within 3 dB as seen in Fig. 1a. This resulted in a rate of side mode suppression (RSMS) of -6.98 dB/nm, defined as the rate at which comb lines not being used for transmission decay. The average relative average noise (RIN) per channel is -133 dB/Hz with an optical linewidth of 3 MHz [5].

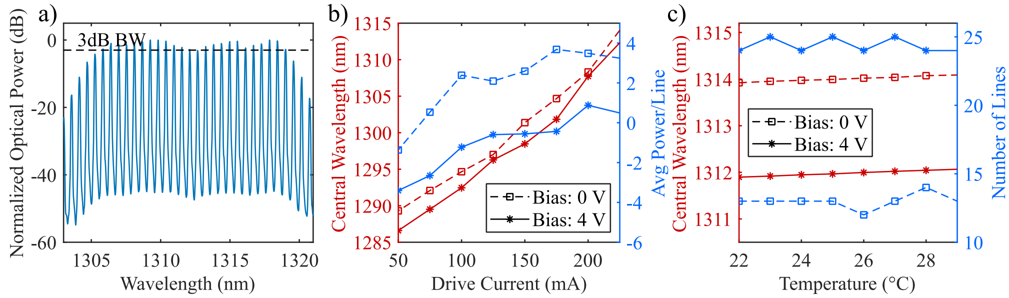

Fig. 1: (a) Normalized spectrum with 25 comb lines at 225 mA, 4 V bias, 25°C. (b) Central wavelength and average power per line variance with driving current. (c) Central wavelength and comb line count within 3 dB change with chip temperature.

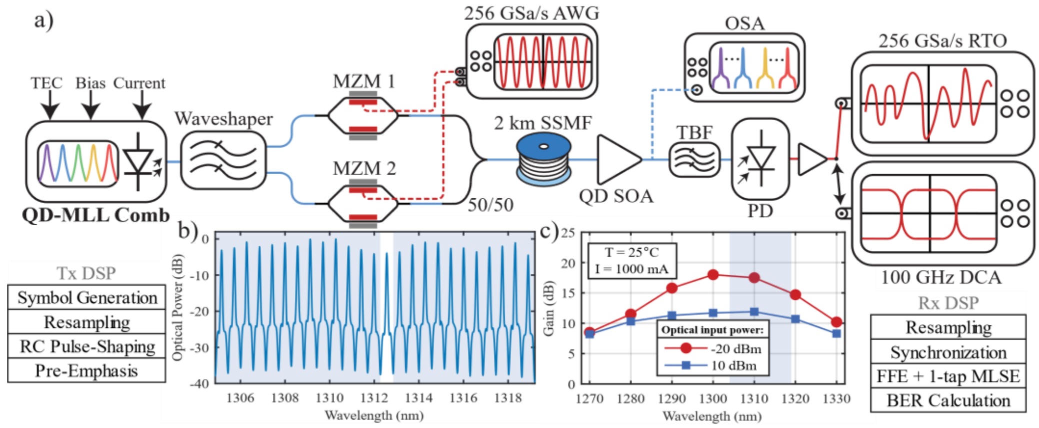

Fig. 2: (a) IM/DD setup with transmitter (Tx) and receiver (Rx) DSP stacks. (b) Optical spectrum after modulation of all channels. The line under test (white background) is modulated separately from the remaining 24 channels (blue background). All channels propagate through fiber and are amplified by one QD-SOA. (c) QD-SOA gain spectrum for -20 and 10 dBm input.

Figure 2 shows the experimental setup. The transmitter source consisted of a QD-MLL connected to an O-band Coherent Waveshaper 4000B programmable optical filter with an insertion loss (IL) of ~5.2 dB. This allowed all channels to be individually tested as the comb line under test (LUT) would be separated from the remaining 24 channels. The LUT is then input into a Hyperlight thin-film lithium niobate (TFLN) Mach-Zehnder modulator (MZM 1). The other 24 channels are directed to a separate modulator (MZM 2) and bulk modulated. The two paths are then combined using a 3 dB coupler. This method is used to emulate a real 25-WDM system. As seen in Fig. 2b, the LUT is iterated through all channels, and all modulated lines always simultaneously propagate through the fibre and optical amplifier. At the transmitter we use a Keysight M8199B 256 GSa/s arbitrary waveform generator (AWG), where 2^{19} PAM symbols are generated. Clipping is applied to optimize the peak to average power ratio. Multi-line experiments use a RC roll-off factor of 0.3. All coded signals are digitally delayed to decorrelate the symbols between MZM 1 and MZM 2. Next, we pre-compensate the frequency response of the AWG and RF link using a digital pre-emphasis filter. We directly drive the two packaged MZMs. The signals are then transmitted over 2 km (and up to 10 km) of single mode fibre (SMF). All channels are then amplified by a single Innolume QD semiconductor optical amplifier (QD-SOA) to compensate for the IL of the WaveShaper, tunable bandpass filter (TBF), and 3dB coupler acting as MUX/DEMUX pairs. The SOA has a max 18 dB gain, 53 nm optical BW, and 20 dBm saturation power. At the receiver, we use an EXFO O-band XTM-50 narrow TBF, with a 5 dB IL to select the LUT during the WDM transmissions. The filter output is connected to a 70 GHz photodiode, an SHF 804B 60 GHz, 22 dB gain RF amplifier and finally to a Keysight 110 GHz, 256 Gsa/s real time oscilloscope (RTO). The ROP, TBF wavelength, and BW are monitored using an optical spectrum analyzer (OSA). The receiver DSP consists of first resampling the signal to 2 samples/symbol and employing a 21-tap feed-forward equalizer (FFE) and a single tap of maximum likelihood sequence estimation (MLSE). Finally, the BER is calculated after the symbols are converted into a bit sequence. The eye diagrams are captured using the same setup connected to a Keysight DCA-X. We note that a fully integrated system solution with custom made multiplexer (MUX) and demultiplexers (DEMUX) pairs would help reduce the dependence on optical amplifiers and stronger DSP. SOAs have already demonstrated strong potential for WDM systems [10, 11] and are needed to overcome some of the power/line output limitations of comb sources when used in short reach transmission systems.

The transmission performance of all comb lines over 2 km can be seen in Fig. 3a. We achieved net 3.2 Tbps (25 × 131 Gbps) using 70 Gbaud PAM-4 under the 6.7% overhead HD-FEC and net 4.18 Tbps (25 × 167 Gbps) using 96 Gbaud PAM-4 under the 14.8% overhead O-FEC. Additionally, net 4.3 Tbps (25 × 172 Gbps) using 80 Gbaud PAM-6 under the 20% overhead SD-FEC and net 5.76 Tbps (25 × 230 Gbps) using 96 Gbaud PAM-8 under the 25% overhead SD-FEC. The edge channels (1305.17 and 1318.9 nm) suffered from worse performance due to the lower power per line. Channels below 1312 nm typically performed better because the SOA provides higher gain at these shorter wavelengths. The comb lines with the highest (line 6, solid) and lowest (line 15, dashed) power were selected to measure the BER sensitivity of the WDM system at different distances, ROPs, and symbol rates as seen in Fig. 3b-d. The highest ROP acquired with line 6 was -6.3 dBm while line 15 only received -8 dBm. The two lines closely follow each other at the same ROP suggesting the system is SNR limited. These results are therefore limited by the high ILs present in the demonstration combined with the low output power/line of the laser which are overcome using an optical amplifier. When all channels are present, these SOAs saturate and thus create an imbalance between the noise of high and low energy symbols. This heightened inter-symbol interference can be mitigated using additional DSP algorithms such as non-linear symbol decision and sequential based equalizers [9]. The increase in BER seen at

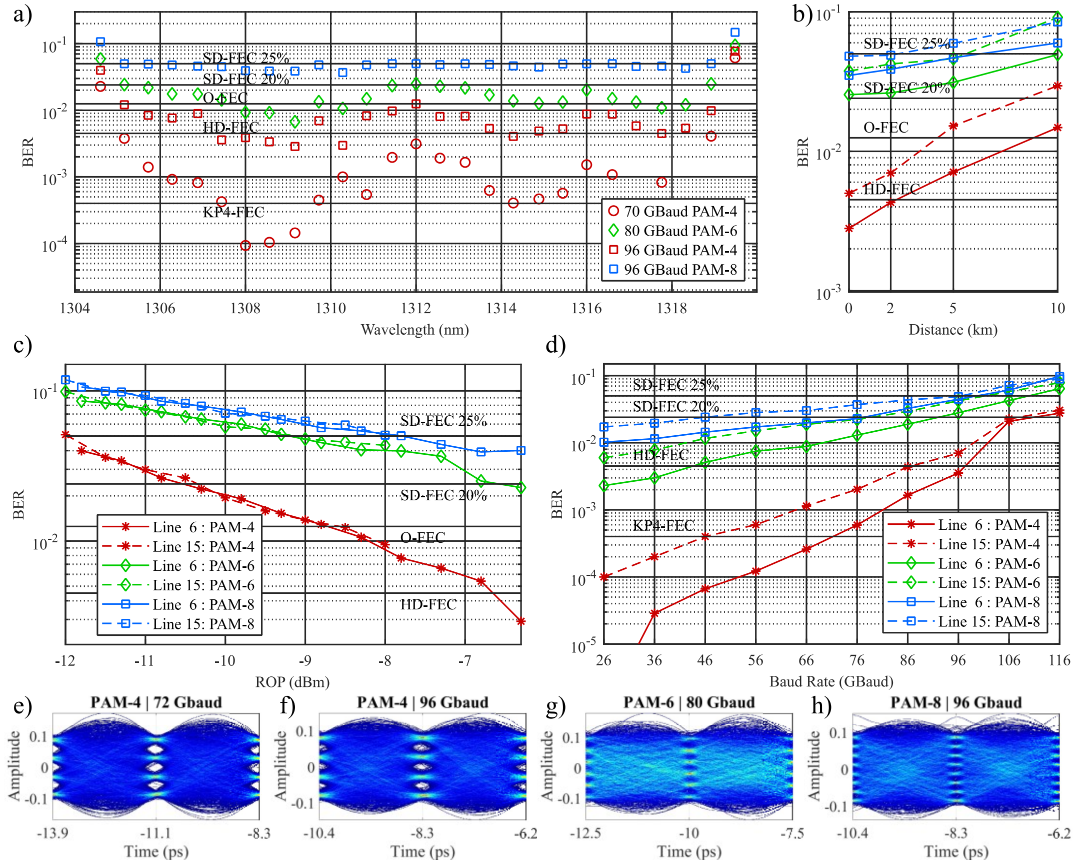

Fig. 3: (a) BER of all tested lines after 2 km SMF. (b) BER versus transmission distance for the best (line 6) and worst (line 15) performing lines. (c-d) BER versus received optical power and symbol rate after 2 km SMF. (e-h) Resulting eye diagrams after 2 km SMF at 72 Gbaud PAM-4, 96 Gbaud PAM-4, 80 Gbaud PAM-6, and 96 Gbaud PAM-8.

higher symbol rates in Fig 3d are due to the spectral overlap between comb channels which can cause severe SNR penalties when using very tight channel spacings. The resulting eye diagrams are plotted in Fig. 3e-h and confirm the capability of QD-MLLs sources to support advanced modulation formats at high symbol rates.

4. Conclusion

We present the first IM/DD transmission system achieving net 3.2, 4.3, and 5.76 Tbps using a single 100 GHz spaced QD-MLL with all channels propagating through 2 km SSMF and one SOA.

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